

Installation of ChampSim Simulator

1. Download the ChampSim simulator from <https://drive.google.com/file/d/1cyIXAd8Ji8BdixSGqsFEdjcH7eQ4e1Cr/view?usp=sharing>
2. Download the SPEC traces from <https://dpc3.compas.cs.stonybrook.edu/champsim-traces/speccpu/>, create a directory named 'dpc3_traces' within the 'ChampSim-master' and copy the traces (.xz files) into it.
3. Before building, give execute permissions to build_champsim.sh and run_champsim.sh files:

```
chmod +x build_champsim.sh  
chmod +x run_champsim.sh
```
4. Additionally, you can also install the necessary packages using the command *sudo apt-get install build-essential*.
5. To build the simulator, use the command `./build_champsim.sh ${BRANCH} ${L1_PREFETCHER} ${L1D_PREFETCHER} ${L2C_PREFETCHER} ${LLC_PREFETCHER} ${LLC_REPLACEMENT} ${NUM_CORE}`
Ex: `./build_champsim.sh bimodal no no no no lru 1` (for 1-core)
`./build_champsim.sh bimodal no no no no lru 4` (for 4-core)
6. On building successfully, a binary file is generated and this binary file is passed as an argument to run the simulator.
7. To run the single-core simulation, use the command `./run_champsim.sh [BINARY] [N_WARM] [N_SIM] [TRACE]`
Ex: `./run_champsim.sh bimodal-no-no-no-no-lru-1core 1 10 400.perlbench-41B.champsimtrace.xz`
8. Similarly, to run the multi-core simulation, use the command `./run_4core.sh [BINARY] [N_WARM] [N_SIM] [N_MIX] [TRACE0] [TRACE1] [TRACE2] [TRACE3]`
Ex: `./run_4core.sh bimodal-no-no-no-no-lru-4core 1 10 0 400.perlbench-41B.champsimtrace.xz`
`401.bzip2-38B.champsimtrace.xz 403.gcc-17B.champsimtrace.xz 410.bwaves-945B.champsimtrace.xz`
9. On successful completion of the simulation, results would be generated in a text file and stored in "results_sim_instr" directory, which contain the values of performance metrics such as Cumulative IPC, Hit Rate etc.

Note: Please ensure that you have installed Linux environment/Ubuntu in your system, as it is needed to run the simulator. If you are facing any issues regarding the installation of the simulator, you can mail me to cs23resch11004@iith.ac.in

Traces to be downloaded

401.bzip2-277B.champsimtrace.xz
403.gcc-16B.champsimtrace.xz
450.soplex-247B.champsimtrace.xz
605.mcf_s-1644B.champsimtrace.xz
459.GemsFDTD-1169B.champsimtrace.xz
619.lbm_s-3766B.champsimtrace.xz
482.sphinx3-1522B.champsimtrace.xz
621.wrf_s-8100B.champsimtrace.xz
623.xalancbmk_s-700B.champsimtrace.xz
628.pop2_s-17B.champsimtrace.xz